

INTERMEDIATE REPORT – **ASSIGNMENT 3**

COMPUTATIONAL BIOFLUID MECHANICS

ACADEMIC YEAR 2025-2026

Name student

Prof. Charlotte Debbaut, Prof. Patrick Segers

Dr. ir. Mohammad Rahimi Gorji, ir. Saar Vermijs, ir. Jessie Duquesne,
ir. Kaat Wille, ir. Irea Vervaeke

ASSIGNMENT 3: SEGMENTATION AND MEASUREMENTS

Make sure that figures/tables are high-quality and complete, and have numbers and clear captions.

Methods: Segmentation and measurements - max. 200 words (/4)

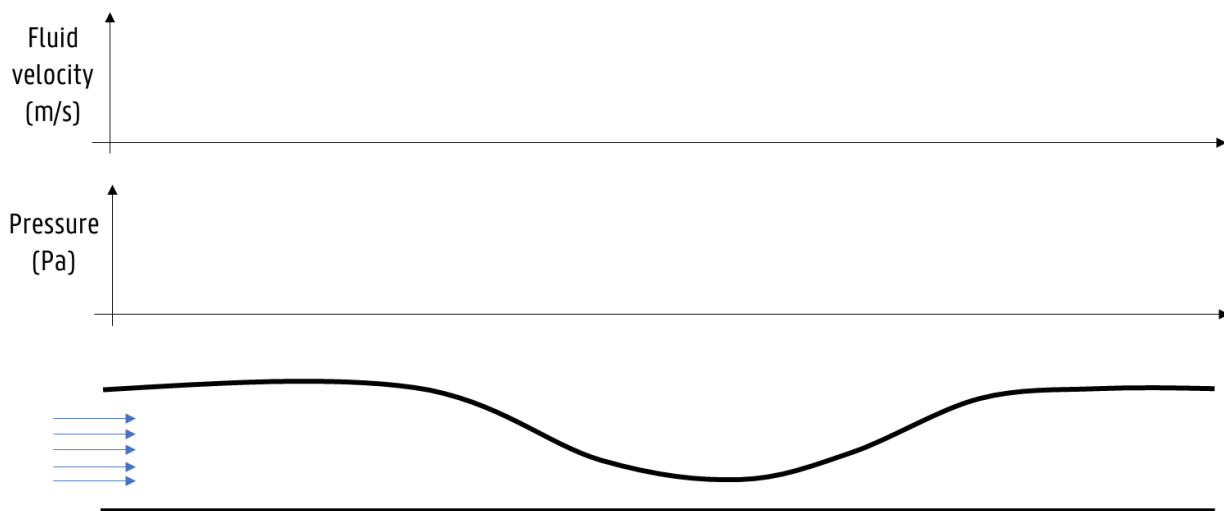
- Describe the segmentation process and include a clear picture of the 3D reconstruction of both the pre-op and post-op case (/2).
- Perform a measurement of the length and minimal diameter of the constricted region in the pre-op case and of diameter in the post-op case (at the same location as in the pre-op case). Clearly indicate these measurements on the 3D reconstructions (/1).
- Calculate the degree (%) of constriction and think wisely about how to do so. Clearly mention and explain the formula used (/1).

(Make sure figures are numbered and have clear captions)

Flow behavior - max. 100 words (/4)

Below you see a sketch of a tube-like structure with an approximately constant diameter and an incoming flow jet, followed by a constricted region, similar to your pre-op case.

- On the graphs, draw (qualitatively) the velocity before, throughout and after the constriction. On which (simple) physical law is this based? (/1)
- What happens with pressure before the constriction, where the diameter of the tube can be considered constant? Do you know an equation to explain this behavior? Which force causes this behavior? (/1)
- What happens with pressure throughout the constriction, knowing already what happens with velocity during the constriction (see above)? Explain on which principle(s) or mathematical equation(s) the pressure/velocity behavior in the constriction is based (/1).
- Knowing what happens to velocity, what happens to pressure after the constriction (/1)?



Sketch of a constricted region with incoming flow jet.

Resistance - max. 50 words (/2)

We want to estimate the resistance of the constriction in your pre-op case. How can you estimate the local resistance of the constriction using CFD results? How would you do this?